

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

SE 19-206-0403 STRENGTH OF MATERIALS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Recognize the fundamental concepts of stress and strain in elastic bodies subjected to external loads.
 CO2: Apply the concept of stress and strain in solving engineering problems.
 CO3: Calculate bending moments, shear forces, axial forces and deformation characteristics of determinate beams.
 CO4: Solve engineering problems related to thin and thick pressure vessels.
 CO5: Solve engineering problems related to circular shafts, helical springs and buckling of columns.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

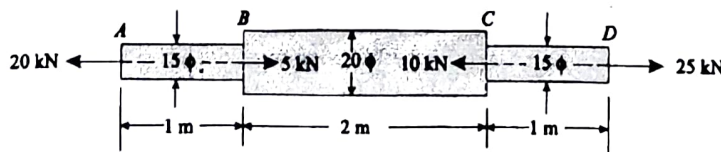
(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Derive an expression for elongation of tapering/conical section.	3	L1	1	1
(b) Using Poisson's ratio, derive the expression for volumetric strain in a 3D stress system.	3	L2	2	1
(c) As a safety engineer explain the application of SFD, BMD and point of contra flexure in load bearing beam.	3	L3	3	2
(d) Explain using a neat sketch the internal stress distribution for a beam subjected to pure bending.	3	L1	3	2
(e) Using an example of a cylindrical bar explain different design criteria with appropriate equations.	3	L4	3	2
(f) How does the slope profile and boundary varies in a beam with (i) simply supported on both ends (ii) fixed support on both ends (iii) cantilever beam.	3	L2	3	1
(g) Using a neat sketch explain the stress and strain in a transmission shaft of an automobile.	3	L4	5	1
(h) Explain the application and assumptions of Euler's theory for struts and columns.	3	L1	5	1

PART B

(4 × 12 = 48)

- II. A steel bar ABCD is subjected to forces as shown in figure, evaluate the force acting on each section using schematic diagram, total elongation of the bar and stress acting on section AB, BC and CD. Take E for steel as 200 GPa. (Φ represents diameter in mm)



OR

(P.T.O.)

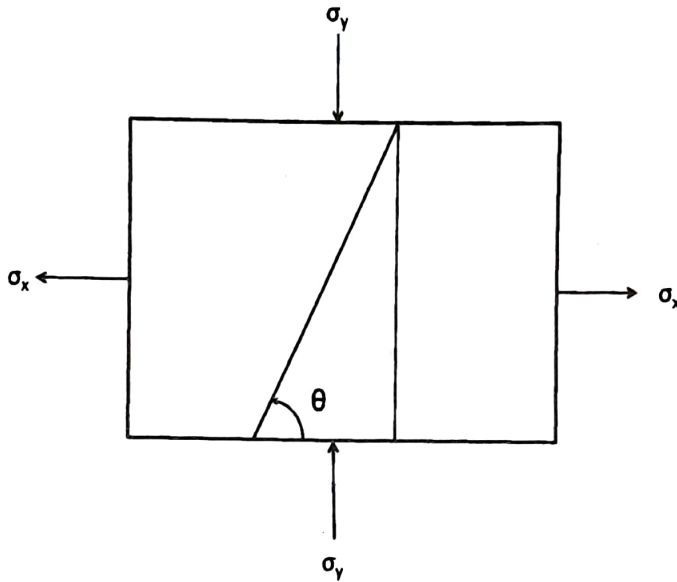
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Marks BL CO

- III. (a) Using an example of compound section of Aluminium and Titanium subjected to an increase in temperature ΔT , explain thermal stress and strain, thermal expansion, tensile and compressive strain and compatibility equations.
 [$\alpha_{\text{aluminium}} = 23.6 \times 10^{-6} / ^\circ\text{C}$ and $\alpha_{\text{titanium}} = 9.4 \times 10^{-6} / ^\circ\text{C}$]
- (b) Using a schematic diagram of a Mohr's circle explain the step by step process to evaluate the normal stress and shear stress at a plane ' θ ' in the following stress system.

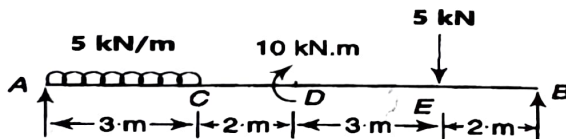
6 L3 2

6 L4 2



- IV. Find the support reactions at A and B and draw SFD and BMD for the following beam (Use appropriate scales for SFD and BMD)

12 L3 3



OR

- V. (a) Derive beam differential equation also explain the relationship connecting shear force bending moment, slope and deflection.
- (b) You were assigned as an industrial visit guide for a group of high school students, one of them asked you why I section beams are widely used in industries, justify your answer theoretically.

8 L1 3

4 L2 3

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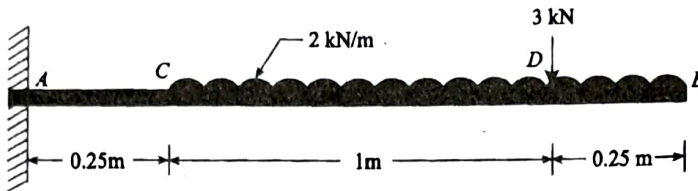
Marks	BL	CO	PO
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VI. (a) Explain the algorithm of double integration method using an example of a simply supported beam with two point load, make suitable assumptions for the magnitude of load and length, take young's modulus of the material as 'E' and moment of inertia of the beam as 'I'.

4	L1	3	1
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(b) Derive the general expression for slope and deflection using McCaulauy's method for the following beam, use young's modulus of the material as 'E' and moment of inertia of the beam as 'I'.

8	L3	3	3
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OR

VII. After an annual inspection of a cylindrical tank in a petroleum industry the safety inspector reported that the change in dimension in the lateral direction is twice that of longitudinal direction, as a safety engineer provide theoretical explanation for this observation. The management also asked to train the inspectors on different internal stress that would develop on a thin cylindrical tank, prepare a note with appropriate sketches.

12	L4	4	4
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VIII. A hollow steel shaft transmit 200 kW of power at 150 rpm. The total angle of twist in a length of 5 m of the shaft is 3° . Find the inner and outer diameter of the shaft if the permissible shear stress is 60 MPa. Take $G = 80 \text{ GPa}$.

12	L3	5	4
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OR

IX. Evaluate the critical load for a strut fixed at both the ends.

12	L1	5	2
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Blooms's Taxonomy Level's

L1 – 27%, L2 – 9%, L3 – 44%, L4 – 20%.
